Jason Liao

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Education

Simon Fraser University — Surrey, BC
BASc Mechatronic Systems Engineering, 3.95/4.33 CGPA — *Anticipated September 2028*

- President's Honour Roll 2024, 2026
- Siemens Mechatronic Systems Certification Program (Systems Assistant, Level 1)

Technical Skills

Mechanical: SolidWorks (FEA and GD&T), Fusion 360 (CAM), FDM 3D printing, DFM, Lathe/Mill/Waterjet
Embedded and Software: C/C++, Python, MATLAB, STM32CubeMX, Git, CAN bus, UART, PWM, FreeRTOS, MS Office Suite, Linux
Industrial Automation: Ladder Logic, Studio 5000 (Logix Designer, Emulate), FactoryTalk View Studio, Siemens TIA Portal

Experience

Automation Designer — May 2025 – December 2025
QCA Systems — *Delta, BC*

- Developed a Studio 5000 Add-On Instruction (AOI) using GNSS coordinates to detect shiploader positions within defined exclusion zones and prevent collisions with a spout change-out tower and adjacent shiploader.
- Wrote Automation Design Documents (ADDs) for Shiploader Slew and Coal Spout PLC programs and HMIs, documenting controls, inputs, alarm banners, interlocks, and process flows.
- Debugged HMI screens for a Stacker-Reclaimer's in FactoryTalk View Studio and produced Software Factory Acceptance Test (FAT) procedures and demonstration videos for client delivery.
- Configured historization for hundreds of PLC I/O and status tags and built Grafana dashboards to monitor equipment position, VFD status, and other operating data.

Projects/Extracurriculars

SFU Ascension Robotics (RoboMaster) | *Mechanical Lead, Embedded Systems* — September 2023 – Present

- Developed an **STM32 CAN bus library** for LingKong BLDC servo motors, implementing packet parsing and commands for velocity, torque, PID parameter setting, and multi-rotation position control.
- Designed, printed, and programmed a self-stabilizing **6-axis leader arm** on an STM32F4 as a controller for a robot's articulated arm; implemented FreeRTOS tasks for encoder feedback, wrist stabilization, and UART communication.
  - Contributed to the team's **2nd-place** finish in the Engineer category at the 2026 ARC Championships.
- Designed a **400+ component robotic turret** in SolidWorks using DFM principles, iteratively prototyping the indexer, pitch mechanism, and hopper with FDM 3D printing to refine component geometry and resolve mechanical issues
- Manufactured the final turret assembly using waterjet-cut fiberglass plates and PPA-CF components, achieving reliable indexing, aiming, and projectile firing at **1500 rounds per minute**
- Wrote a **6-DOF inverse kinematics solver** in MATLAB using the Robotics System Toolbox and the robot arm's URDF to validate joint configurations, workspace, and reachability for inserting a cube into a narrow slot.

SubVision (MATE ROV) | *Mechanical team* — September 2024 – May 2025

- Prototyped and 3D-printed a 2-DOF **four-bar linkage gripper** using SolidWorks, enabling an ROV to manipulate a PVC pipe underwater.

Sip-and-Puff Fishing Rod — January 2024 – April 2024

- Designed the CAD model and electrical controls for a sip-and-puff fishing rod, enabling a client with limited upper-body mobility to cast, reel, raise, and lower the rod using only mouth gestures.